

- 1) A winery wants to develop a model to predict the quality of wine based on its chemical properties. Given a dataset containing features such as alcohol content, pH level, acidity, sugar concentration, sulfur dioxide levels, and density, the goal is to predict the wine quality score (on a scale of 0 to 10). (**WineQuality**)
- 2) A marketing company wants to analyze customer behavior to determine whether a person purchased a product after seeing a social media ad. Given a dataset containing customer features such as age, gender, and annual income, the goal is to predict whether the customer purchased the product (Yes/No) or (1/0). (**Social_media_ads**)_use **Naive Bayes Classifier/ svm classifier**.
- 3) A marketing company wants to analyze customer behavior to determine whether a person purchased a product after seeing a social media ad. Given a dataset containing customer features such as age, gender, and annual income, the goal is to predict whether the customer purchased the product (Yes/No) or (1/0). (**Social_media_ads**)_use **SVM Classifier/ KNN classifier**.
- 4) A company wants to develop a model to predict the salary of employees based on various factors such as age, gender, education level, job title, and years of experience. Given a dataset containing these employee attributes, the goal is to build a predictive model that estimates the expected salary (in dollars) for a given individual. (**salary_data**)
- 5) An online retail company wants to segment its customers based on their purchasing behavior to optimize marketing strategies and improve

customer experience. Given a dataset the goal is to group customers into distinct clusters based on their purchasing patterns. **(online_retail)**

- 6) An e-commerce company wants to predict whether a customer will complete a purchase based on their browsing behavior. The dataset contains features such as page visits, session duration, bounce rates, traffic type, visitor type, and special promotions. The goal is to build a classification model that predicts whether a visitor will generate revenue (True/False), helping the company optimize marketing strategies and improve conversion rates. **(online_shoppers_intention)_Use Naive Bayes Classifier/ knn classifier.**
- 7) An e-commerce company wants to predict whether a customer will complete a purchase based on their browsing behavior. The dataset contains features such as page visits, session duration, bounce rates, traffic type, visitor type, and special promotions. The goal is to build a classification model that predicts whether a visitor will generate revenue (True/False), helping the company optimize marketing strategies and improve conversion rates. **(online_shoppers_intention)_Use knn classifier.**
- 8) A healthcare organization wants to develop a predictive model to detect lung cancer in patients based on various medical and lifestyle factors. The goal is to predict whether a patient has lung cancer (Yes/No). **(Lungcancer)_Use knn classifier.**
- 9) A healthcare organization wants to develop a predictive model to detect lung cancer in patients based on various medical and lifestyle factors. The goal is to predict whether a patient has lung cancer (Yes/No). **(Lungcancer)_Use knn classifier**

- 10) A real estate company wants to segment houses based on their furnishing status to better understand market trends and customer preferences. Given a dataset, the goal is to group houses into clusters based on similarities in their characteristics. **(Housing)**
- 11) A power generation company wants to predict the energy yield (TEY) of gas turbines based on various environmental and operational parameters. Given a dataset containing features such as ambient temperature, air pressure, air humidity, fuel flow, turbine inlet temperature, and exhaust temperature, the goal is to build a regression model that accurately predicts the turbine's energy output (TEY in MW). **(Gas_turbine)**
- 12) A financial firm wants to develop a stock price prediction model to assist investors in making informed trading decisions. Given a dataset containing historical stock data such as opening price, highest price, lowest price, closing price, trade volume, and turnover, the goal is to build a regression model that accurately predicts the future closing price of the stock. **(Glaxo)**
- 13) A financial institution wants to develop a fraud detection system to classify individuals based on their risk level for financial fraud. Given a dataset, the goal is to build a classification model that predicts whether an individual is at high or low risk of financial fraud based on their taxable income and demographic details. **(Fraud_detection)Use svm classifier.**
- 14) A financial institution wants to develop a fraud detection system to classify individuals based on their risk level for financial fraud. Given a

dataset, the goal is to build a classification model that predicts whether an individual is at high or low risk of financial fraud based on their taxable income and demographic details. **(Fraud_detection) Use svm classifier.**

- 15) A forest management agency wants to develop a forest fire severity prediction model to estimate the burned area in case of a fire outbreak. Given a dataset, the goal is to build a regression model that predicts the burned area of a forest fire. **(forestfires)**
- 16) A forest management organization wants to build a fire severity classification model to categorize forest fires as either "small" or "large". Given historical fire data, the goal is to train a classification model that can predict whether a fire will be small or large. **(forest)**
Use knn classifier.
- 17) A forest management organization wants to build a fire severity classification model to categorize forest fires as either "small" or "large". Given historical fire data, the goal is to train a classification model that can predict whether a fire will be small or large. **(forest)**
Use knn classifier.
- 18) How can a Support Vector Machine (SVM) be used to classify patients as diabetic or non-diabetic based on medical attributes such as glucose level, blood pressure, insulin levels, BMI, age, and family history? What is the optimal choice of kernel (linear, polynomial, or RBF) for improving classification accuracy in this healthcare prediction task? **(diabetes_pima).**

- 19) How can a knn classifier be used to classify patients as diabetic or non-diabetic based on medical attributes such as glucose level, blood pressure, insulin levels, BMI, age, and family history? **(diabetes_pima).**
- 20) A law enforcement agency wants to develop a crime rate prediction model to estimate the crime rate in different regions. Given a dataset containing socioeconomic factors, the goal is to train a regression model that predicts the crime rate in a given area. **(CrimeDataset).**
- 21) A government agency wants to develop an income classification model to predict whether an individual earns more or less than a certain income threshold. Given a dataset containing demographic and employment-related attributes such as age, workclass, education level, occupation, marital status, and hours worked per week, the goal is to train a classification model that can categorize individuals into two income groups. **(adult) Use knn classifier.**
- 22) A government agency wants to develop an income classification model to predict whether an individual earns more or less than a certain income threshold. Given a dataset containing demographic and employment-related attributes such as age, workclass, education level, occupation, marital status, and hours worked per week, the goal is to train a classification model that can categorize individuals into two income groups. **(adult) Use svm classifier.**
- 23) How can a knn classifier be used to predict whether a customer will subscribe to a term deposit based on demographic and financial attributes such as age, job type, education, balance, loan status, and

previous marketing campaign interactions? How does the assumption of feature independence affect the model's performance in this financial classification task? **(bank-full)**.

- 24) How can a SVM classifier be used to predict whether a customer will subscribe to a term deposit based on demographic and financial attributes such as age, job type, education, balance, loan status, and previous marketing campaign interactions? How does the assumption of feature independence affect the model's performance in this financial classification task? **(bank-full)**
- 25) A medical research institute wants to develop an AI-based diagnostic model to classify brain tumors into Benign or Malignant based on patient characteristics such as age, gender, tumor size, location, histology, stage, symptoms, and MRI results. **(brain_tumor)_Use knn classifier.**
- 26) A medical research institute wants to develop an AI-based diagnostic model to classify brain tumors into Benign or Malignant based on patient characteristics such as age, gender, tumor size, location, histology, stage, symptoms, and MRI results. **(brain_tumor)_Use svm classifier.**
- 27) How can a Support Vector Machine (SVM) model be used to classify breast cancer tumors as benign or malignant based on tumor characteristics such as radius, texture, perimeter, area, and smoothness? What is the optimal kernel function for achieving high classification accuracy? **(breast_cancer)_Use svm classifier.**
- 28) How can a Support Vector Machine (SVM) model be used to classify breast cancer tumors as benign or malignant based on tumor

characteristics such as radius, texture, perimeter, area, and smoothness? What is the optimal kernel function for achieving high classification accuracy? **(breast_cancer)_Use naïve bayes classifier.**

- 29) Develop a predictive model to estimate the profit of a startup based on its R&D expenditure, administration costs, and marketing spending. **(50_startups)**
- 30) How can data visualization techniques be used to analyze the performance and efficiency of different irrigation machines? What trends or patterns can be identified in water usage, coverage area, and operational efficiency across different machine types? **(irrigation_machine)**
- 31) Using a Naive Bayes classification model, predict whether an employee is likely to leave the company (attrition) based on numerical features such as age, daily rate, distance from home, years at company, and other continuous employee-related metrics. **(employee_attrition)**
- 32) Develop a regression model to predict the valuation of a startup based on its industry, country, and valuation date. **(Startups)**
- 33) Using a Naive Bayes classification model, predict whether a Grammy-nominated song, album, or artist will win an award based on features such as award category, artist details, song/album title, and year. **(grammy_winners)**

- 34) Predict the selling price of a home based on property characteristics such as size, number of rooms, location, floor level, and heating type. **(home_price)**
- 35) Using a Support Vector Machine (SVM) classifier, predict whether a laptop review is positive or negative based on the review text, star rating, and other associated metadata such as product name and review title. **(laptops_dataset)**
- 36) Using a Naive Bayes classifier, predict whether a song will become a high-streaming hit based on its musical features (such as BPM, key, mode, danceability, energy, and acousticness) and metadata like release date and playlist visibility. **(spotify_2023)**
- 37) Using a Support Vector Machine (SVM) classifier, predict the sentiment (positive or negative) of a Yelp review based on its textual content and user feedback metrics such as 'useful', 'cool', and 'funny' votes. **(yelp)**
- 38) Develop a predictive model that estimates an individual's anxiety level (on a scale from 1 to 10) based on demographic, lifestyle, physiological, and psychological factors such as age, sleep hours, stress levels, physical activity, caffeine/alcohol consumption, and therapy history. **(family_anxiety_14_dataset)**
- 39) Develop a regression model to predict the total flight fare (in BDT) based on features such as base fare, duration, class, booking source, seasonality, and number of days before departure. **(Flight_Price_Dataset_of_Bangladesh)**

- 40) Build a Naive Bayes classifier to determine whether a banknote is genuine or counterfeit based on its statistical properties such as variance, skewness, curtosis, and entropy extracted from image analysis.
(bank_note_data)